SECTIONS OF AN ETALE SHEAF ARE UNCHANGED BY BASE CHANGE ALONG A SEPARABLE ALGEBRAIC CLOSURE OF THE BASE FIELD

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument: both hypotheses of the cited sections-upstairs result are verified in the proof, including the geometric fibre computation with its named algebraic input.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Sections computed upstairs. Same chapter, Lemma lemma-sections-upstairs: for a flat quasi-compact base change with connected geometric fibres the sections of a sheaf and of its pullback agree. Supplies the criterion verified in the proof.
- Irreducibility of the tensor product of fields over a separably closed field. Algebra, Lemma algebra-lemma-separably-closed-irreducible. Supplies the connectedness of the geometric fibres of p.

Source reference. Original lemma 0A3H: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 5030–5030. Navigation pointer invoked at proof line 5152; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00I7: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 5157–5157. Navigation pointer invoked at proof line 5157; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let K/k be an extension of fields with k separably algebraically closed. Let S be a scheme over k. Denote $p : S_K = S \times_{\mathrm{Spec}(k)} \mathrm{Spec}(K) \to S$ the projection. Let $\mathcal{F}$ be a sheaf on $S_{étale}$. Then $\Gamma(S, \mathcal{F}) = \Gamma(S_K, p_{small}^{-1}\mathcal{F})$.*

Proof. Follows from Lemma 0A3H. Namely, it is clear that p is flat and quasi-compact as the base change of $\mathrm{Spec}(K) \to \mathrm{Spec}(k)$. On the other hand, if $\overline{s} : \mathrm{Spec}(L) \to S$ is a geometric point, then the fibre of p over $\overline{s}$ is the spectrum of $K \otimes_k L$ which is irreducible hence connected by Algebra, Lemma 00I7. □